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Računarski vid

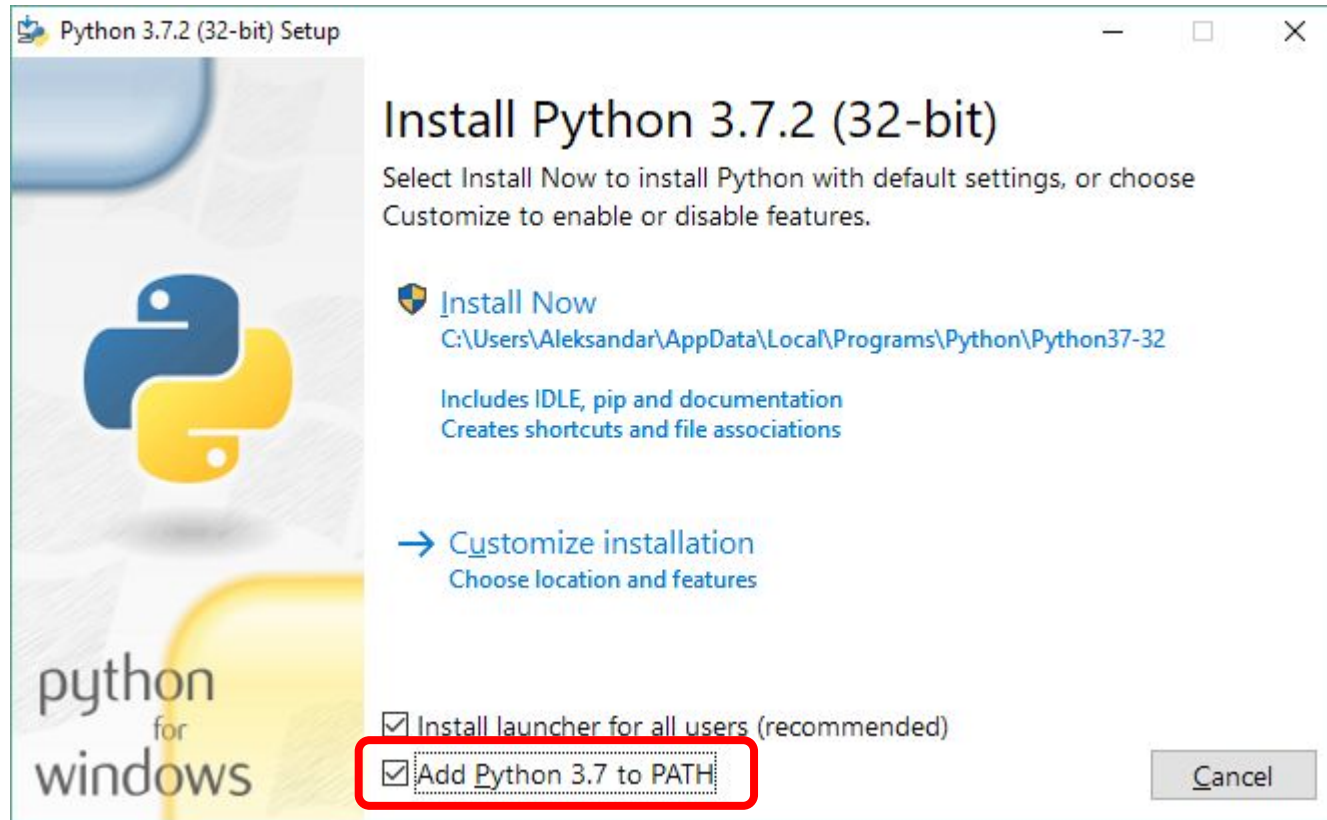
Computer Vision (CV)

Python i biblioteke
Numpy, OpenCV i
Matplotlib

Instaliranje Python-a i PyCharm-a

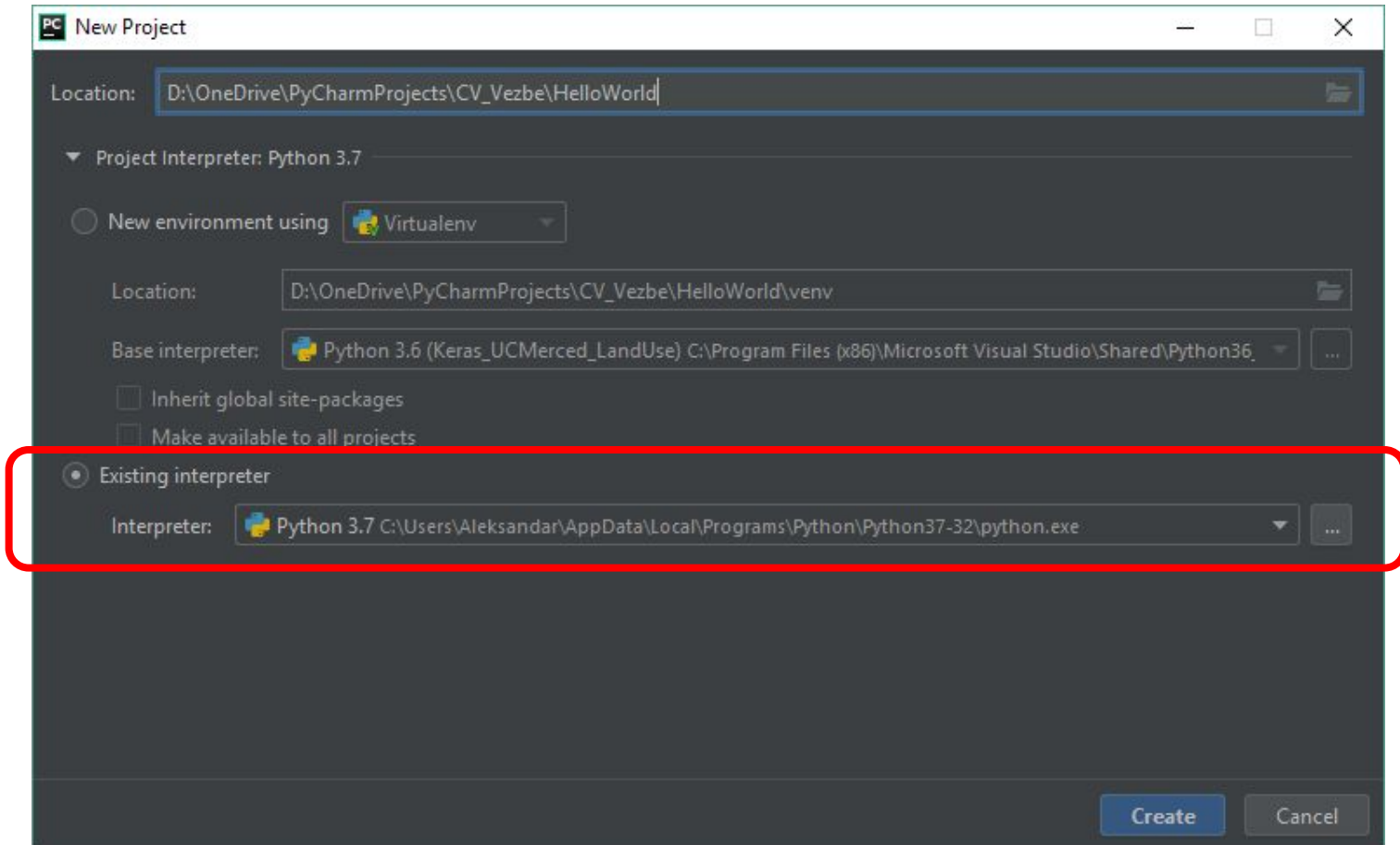
- Instalacija Python-a:
 - <https://www.python.org/downloads/>
- Instalacija PyCharm okruženja za razvoj:
 - <https://www.jetbrains.com/pycharm/download/>
- Za detaljnije instrukcije pogledati video:
 - <https://www.youtube.com/watch?v=rfscVS0vtbw>

Instaliranje Python-a

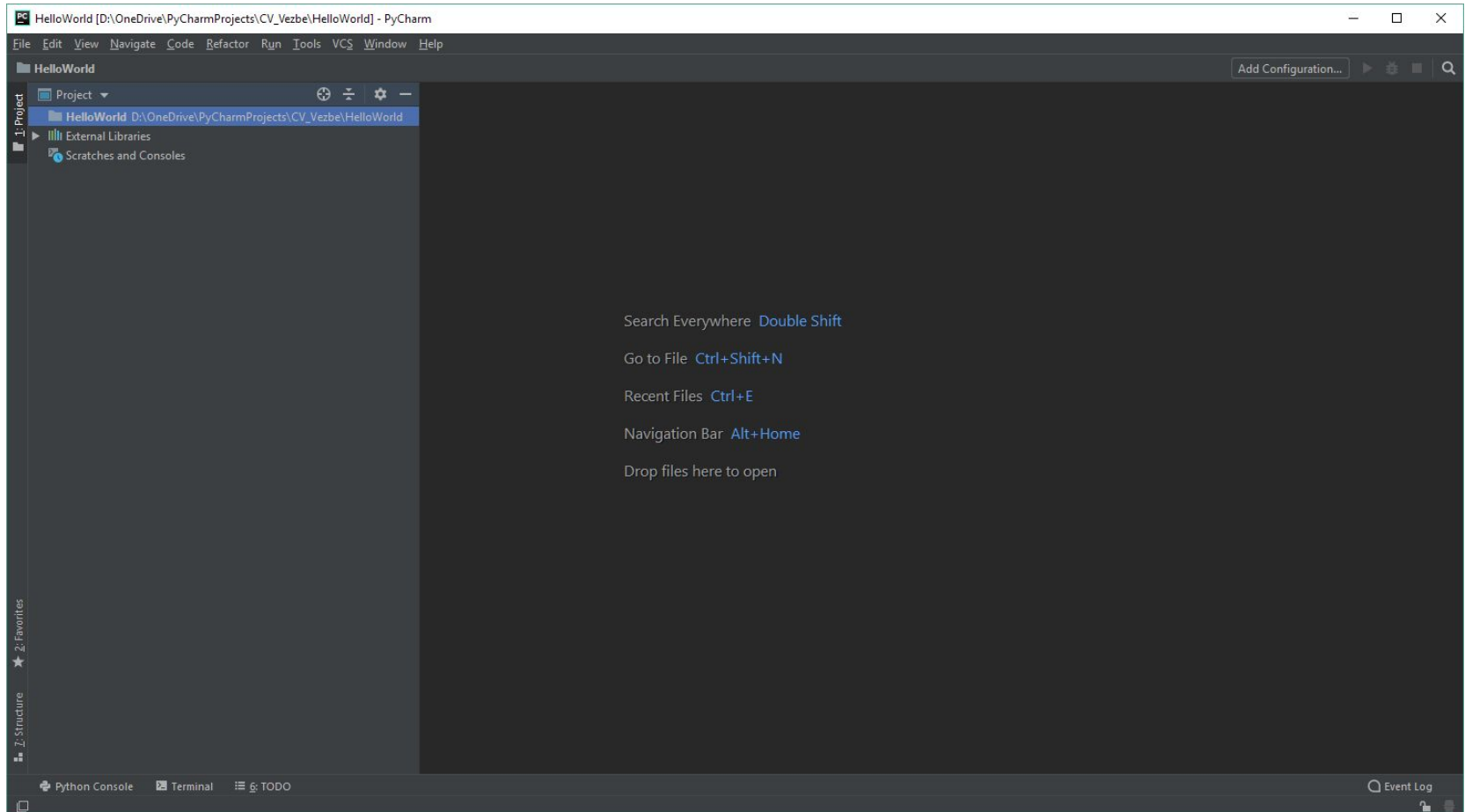


BOLJE INSTALIRATI x64 VERZIJU JER TensorFlow RADI SAMO SA NJOM

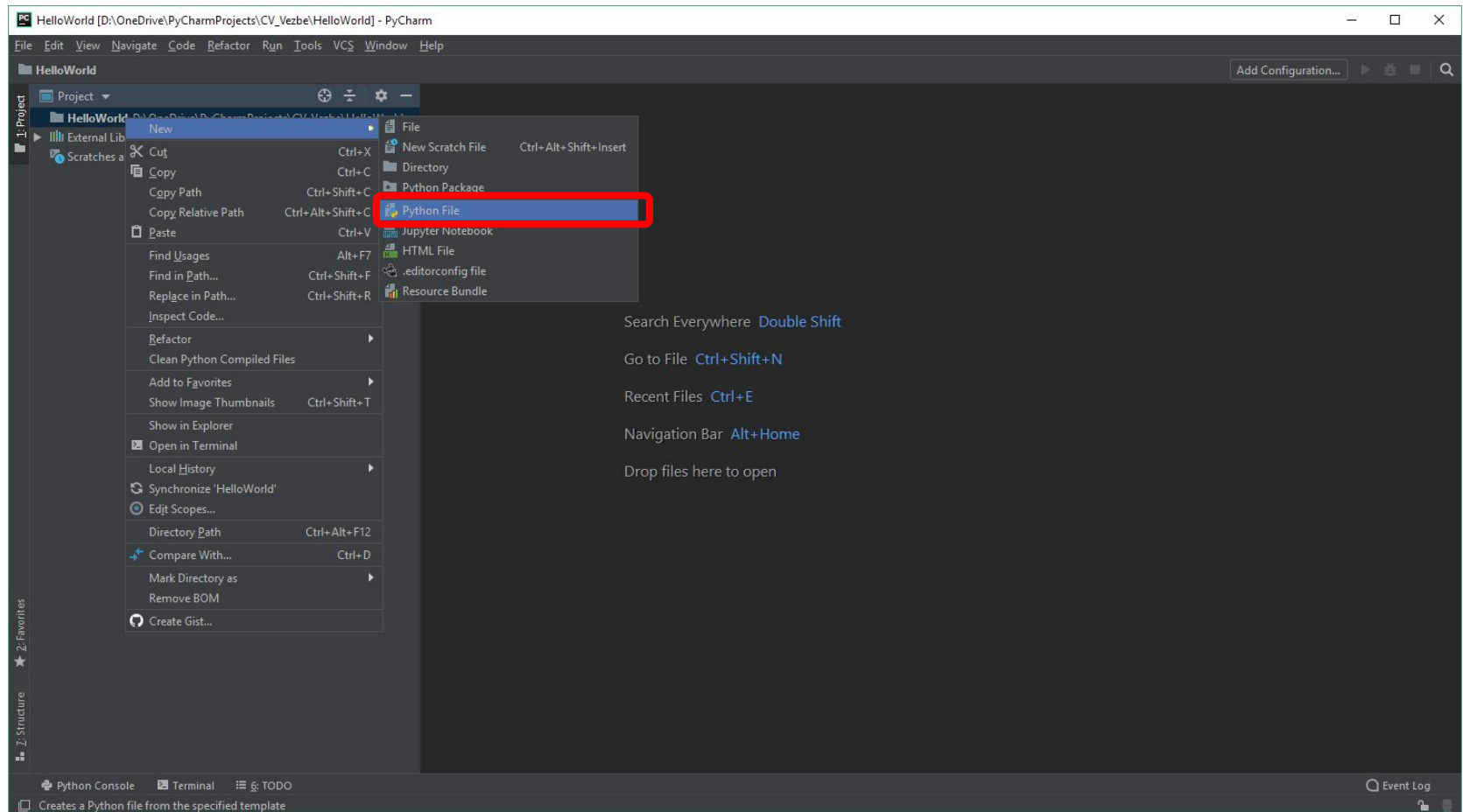
Kreiranje projekta u PyCharm-u



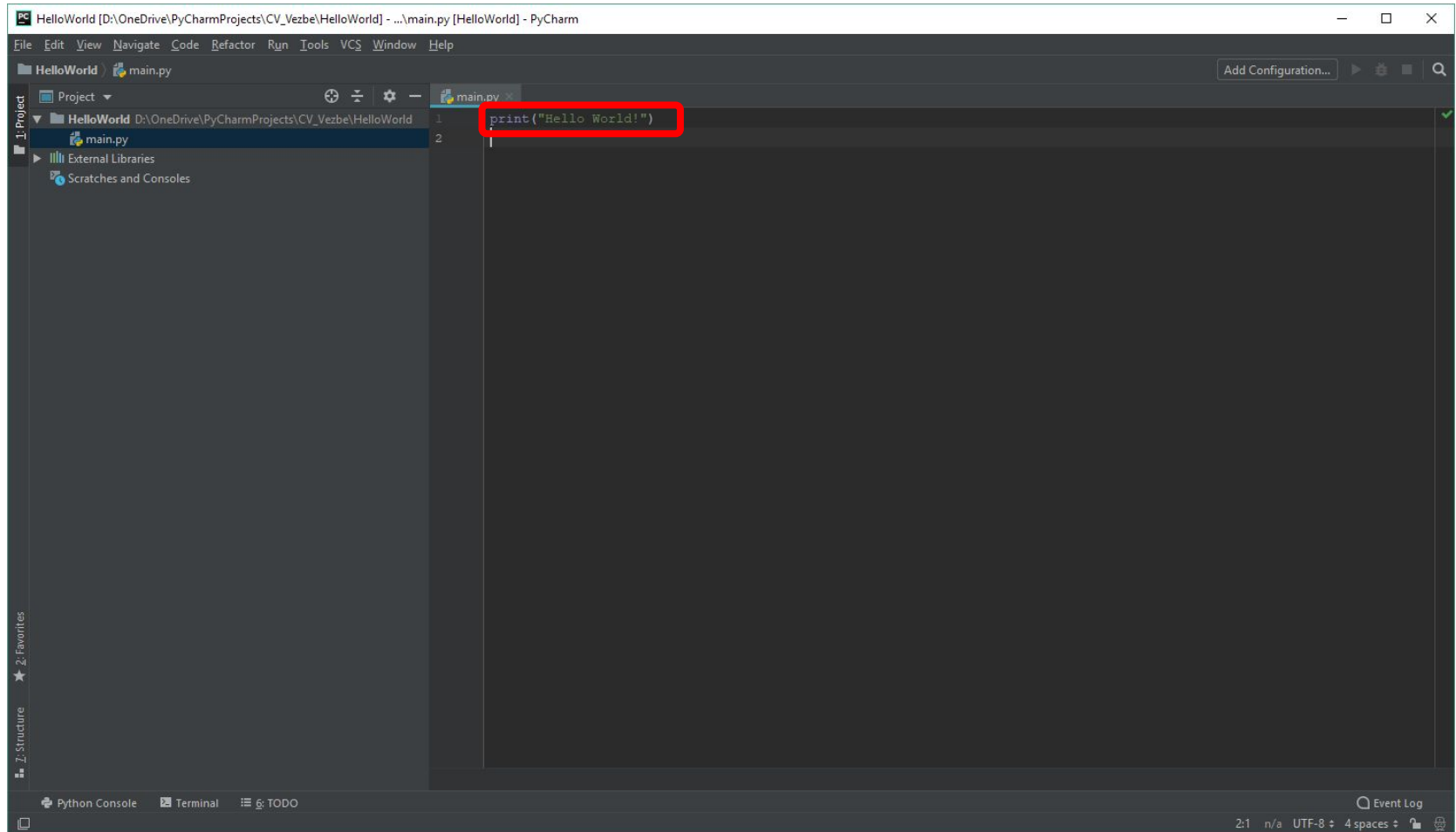
Izgled PyCharm okruženja



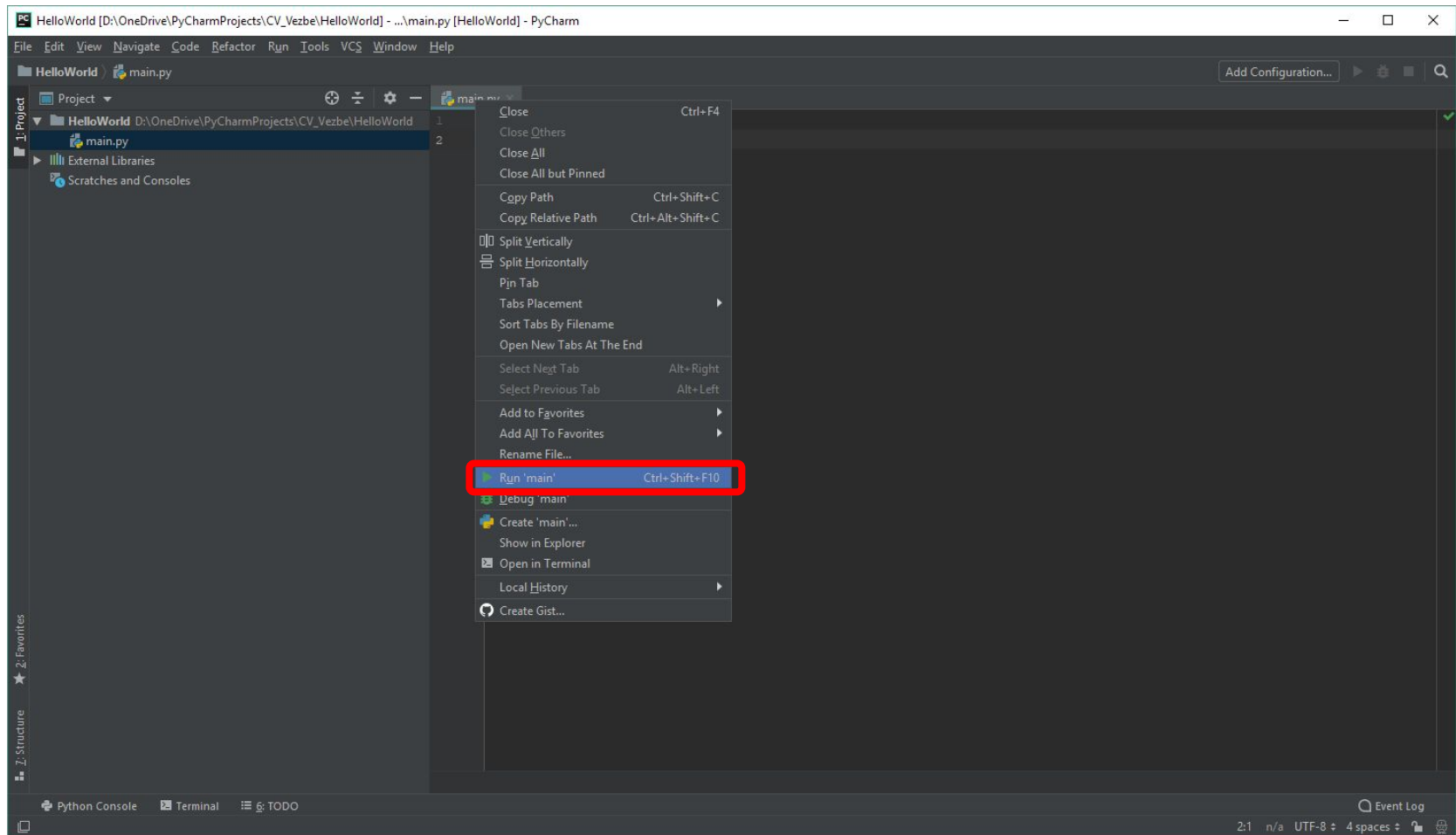
Dodavanje Python fajla u projekat



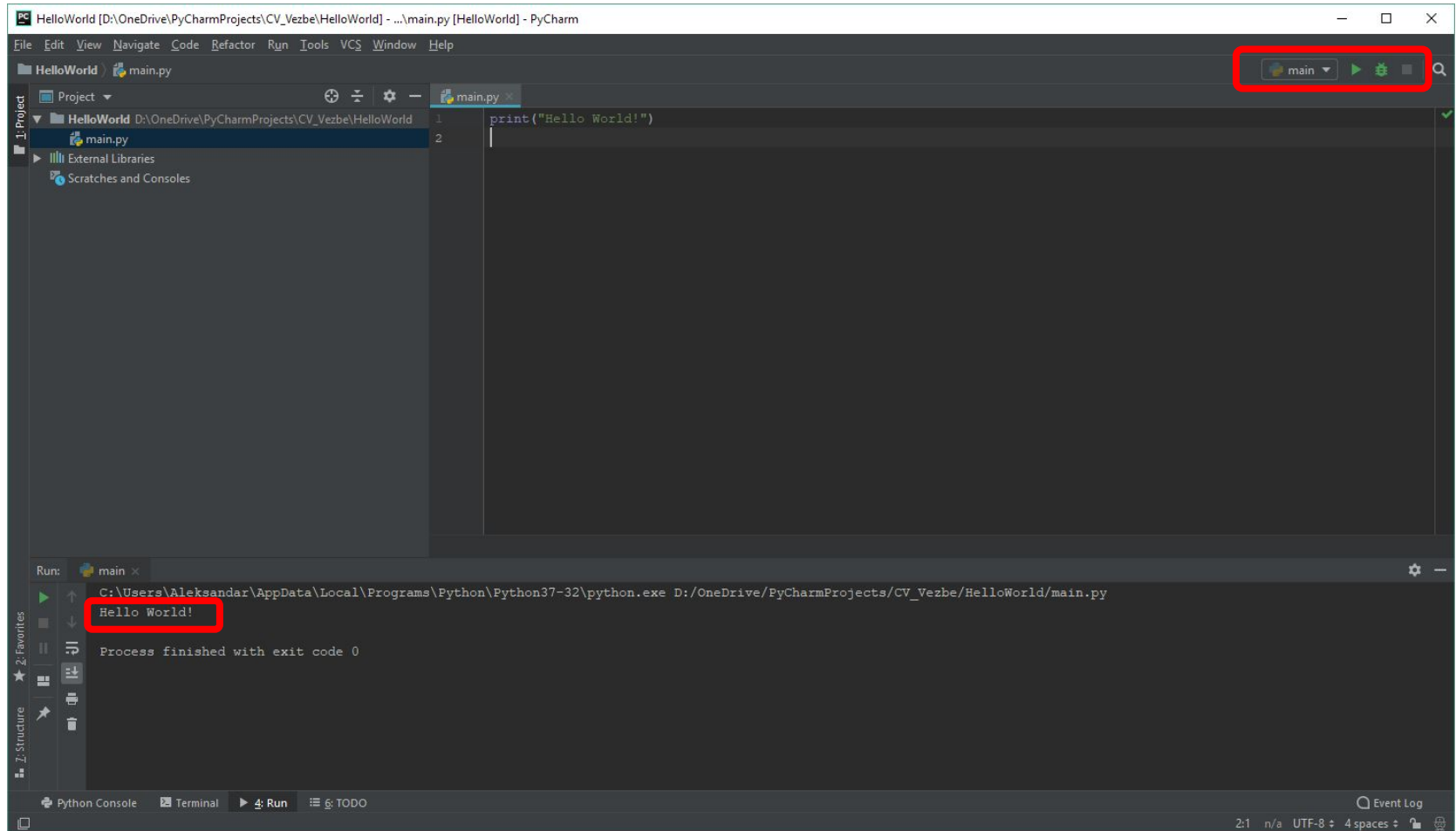
Hello World primer



Startovanje programa



Rezultat izvršenja programa



Instaliranje OpenCV-a preko pip-a

The screenshot shows the PyCharm IDE interface. The top toolbar contains icons for running, debugging, and searching. The left sidebar shows the project structure with 'HelloWorld' and 'main.py'. The main editor displays the code for 'main.py', which contains a single line: `print("Hello World!")`. The bottom panel is divided into two tabs: 'Python Console' and 'Terminal'. The 'Terminal' tab is active and shows the command `pip3 install --upgrade opencv-contrib-python` being executed. The output of the command is visible, showing the download progress of the package. The 'Terminal' tab is highlighted with a red box and the number '1' below it. The command line itself is also highlighted with a red box and the number '2' above it.

```
Terminal: Local x +
Microsoft Windows [Version 10.0.17134.590]
(c) 2018 Microsoft Corporation. All rights reserved.

D:\OneDrive\PyCharmProjects\CV_Vezbe\HelloWorld> pip3 install --upgrade opencv-contrib-python
Collecting opencv-contrib-python
  Downloading https://files.pythonhosted.org/packages/8e/2a/7ac92ff2e605bda7f827a5ff28a9fa16f26fefbb8af9185134290850c458/opencv\_contrib\_python-4.0.0.21-cp37-cp37m-win32.whl (27.2 MB)
100% |-----| 27.2MB 1.1MB/s
```

1

Verzija OpenCV-a

- Kod novijih verzija ove biblioteke ne funkcioniše autocomplete
- Komanda koju je potrebno ukucati u terminal je sledeća:
 - `pip install opencv-contrib-python==4.5.5.62`

OpenCV Hello World

```
import numpy as np
import cv2 as cv

cap = cv.VideoCapture(0)
while True:
    retVal, img = cap.read()
    img = cv.cvtColor(img, cv.COLOR_RGB2GRAY)
    img = cv.medianBlur(img, 7)

    cv.imshow('Camera', img)
    if cv.waitKey(1) & 0xFF == ord('q'):
        break

cap.release()
cv.destroyAllWindows()
```

Python tutorijal

- <https://docs.python.org/3/tutorial/>
- Interpreterski jezik
- Kompaktan kod
- Lako se uči
- Puno biblioteka ga čini izuzetno moćnim
- Nazvan po Letećem cirkusu Montija Pajtona

Promenljive i tipovi podataka

```
op = "Zbir"  
a = 5  
b = 7  
c = a + b  
print("{} brojeva {} i {} je {}".format(op, a, b, c))
```

```
text = "Tekst"  
integer = 10  
number = 2.5 # double-precision floating point number  
boolean = True # or False  
complex_number = 5+3j
```

Rad sa brojevima

```
n = 5
m = 2
t = n / m      # 2.5 (deljenje integera vraca float)
k = n // m     # 2 (celobrojno deljenje)
r = n % m      # 1 (ostatak pri deljenju)
q = n ** m     # 25 (stepenovanje)
p = round(t)   # 2 (zaokruzivanje)
print(n, m, t, k, r, q, p)
```

Rad sa stringovima

```
a = "Prvi red, 'abc'"
b = "Drugi red, \"abc\""
c = 'Treci red, "abc"'
d = a + '\n' + b + "\n" + c
print(d)
```

```
m = input("Unesite broj: ")
print(m, type(m))
k = int(m)
print(k, type(k))
```

```
n = 5
t = "Broj " + str(n)
print(len(t))           # 6
print(t)                # Broj 5
print(t[0], type(t[0])) # B <class 'str'>
print(t[1:3])           # ro
print(t[:2])            # Br
print(t[2:])            # oj 5
print(t[-3:])           # j 5
```


Liste

```
kv = [1, 4, 9, 16, 25]
print(kv)          # [1, 4, 9, 16, 25]
print(kv[0])       # 1
print(kv[-1])      # 25
print(kv[-3:])     # [9, 16, 25]
print(kv[:])       # [1, 4, 9, 16, 25]
kv = kv + [36, 49, 64, 81] # dodavanje liste
kv.append(10**2)        # dodavanje elementna
print(kv)               # [1, 4, 9, 16, 25, 36, 49, 64, 81, 100]
kv[-3:] = []           # brise zadnja tri elementa
print(kv)               # [1, 4, 9, 16, 25, 36, 49]
print(len(kv))          # 7

m1 = [['a', 'b', 'c'],
      [1, 2, 3]]
print(m1[0][1])        # b
print(m1[1][:])         # [1, 2, 3]
```

Višestruka dodela (fibonačijev niz)

```
a, b = 0, 1
while a < 1000:
    print(a, end=', ')
    a, b = b, a+b
```

Kontrola toka (*if* naredba)

```
x = int(input("Unesite celi broj: "))
if x < 0:
    x = 0
    print('Negativan broj korigovan na nulu')
elif x == 0:
    print('Nula')
else:
    print('Pozitivan broj')
```

```
if x is None:
    pass
if x is not None:
    pass
if x == 0 and (y is None or y == ''):
    pass

for i in range(10):
    if not(i >= 3 and i <= 6):
        print(i, end=' ') # 0 1 2 7 8 9
```

Kontrola toka (*for* naredba)

```
reci = ['knjiga', 'lopta', 'pas']
for rec in reci:
    print(rec, len(rec))
# knjiga 6
# lopta 5
# pas 3

for i in range(len(reci)):
    print(i, reci[i])
# 0 knjiga
# 1 lopta
# 2 pas

for i in range(2, 10, 3):
    print(i, end=' ')
# 2 5 8
```

Funkcije

```
def fib(n=100):  
    lf = []  
    a, b = 0, 1  
    while a < n:  
        lf.append(a)  
        a, b = b, a+b  
    return lf  
  
print(fib())  
# [0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89]  
print(fib(50))  
# [0, 1, 1, 2, 3, 5, 8, 13, 21, 34]
```

Funkcije (argumenti)

```
def ask_ok(prompt, retries=4, reminder='Please try again!'):
    while True:
        ok = input(prompt)
        if ok in ('y', 'ye', 'yes'):
            return True
        if ok in ('n', 'no', 'nop', 'nope'):
            return False
        retries = retries - 1
        if retries < 0:
            raise ValueError('invalid user response')
        print(reminder)
```

```
ask_ok('Do you really want to quit? ')
ask_ok('OK to overwrite the file? ', 2)
ask_ok('OK to overwrite the file? ',
        reminder='Come on, only yes or no!')
```

Lamda izrazi

```
pairs = [(1, 'one'), (2, 'two'),  
         (3, 'three'), (4, 'four')]  
pairs.sort(key=lambda pair: pair[1])  
print(pairs)  
# [(4, 'four'), (1, 'one'), (3, 'three'), (2, 'two')]
```

```
def op(sign):  
    if sign == '+':  
        return lambda a, b: a + b  
    elif sign == '-':  
        return lambda a, b: a - b  
    raise ValueError('Invalid operation')
```

```
plus = op('+')  
minus = op('-')  
print(plus(5, 3), minus(5, 3)) # 8 2
```

Operacije nad listama

```
ls = list()           # kreira praznu listu
ls.append(-1)         # dodaje element na kraj liste
print(ls)             # [-1]
ls.extend(range(3))   # dodaje elemente na kraj liste na bazi iteratora
print(ls)             # [-1, 0, 1, 2]
ls.insert(1, 0)       # dodaje element 0 na indeksnu poziciju 1
print(ls)             # [-1, 0, 0, 1, 2]
ls.remove(0)          # izbacuje prvo pojavljivanje elementa
print(ls)             # [-1, 0, 1, 2]
print(ls.pop(), ls)   # 2 [-1, 0, 1]
print(ls.index(0))    # 1

lsr = ls.copy()       # kreira i vraca kopiju liste
lsr.reverse()         # invertuje redosled elemenata
print(ls, lsr)        # [-1, 0, 1] [1, 0, -1]

print(['a', 'b', 'b', 'c'].count('b')) # 2 (broj pojavljivanja elementa)

kv = list(map(lambda x: x**2, range(1, 6))) # kreira listu pomocu map funkcije
print(kv) # [1, 4, 9, 16, 25]
kv = [x**2 for x in range(1, 6)] # kreira listu pomocu for naredbe
print(kv) # [1, 4, 9, 16, 25]

lt = [(x, y) for x in [1, 2, 3] for y in [3, 1, 4] if x != y]
print(lt) # [(1, 3), (1, 4), (2, 3), (2, 1), (2, 4), (3, 1), (3, 4)]
```


N-torke (tuples)

```
t = 12345, 54321, 'hello!' # inicijalizacija
print(t, t[0]) # (12345, 54321, 'hello!') 12345

u = t, (1, 2, 3, 4, 5) # ugnjezdavanje n-torki
print(u) # ((12345, 54321, 'hello!'), (1, 2, 3, 4, 5))

empty = ()
singleton = 'hello', # obratiti paznju na zarez na kraju
print(len(empty), empty) # 0 ()
print(len(singleton), singleton) # 1 ('hello',)

singleton[0] = 'greska' # baca izuzetak
# TypeError: 'tuple' object does not support item assignment

t = (10, 20, 30)
x, y, z = t # raspakovanje vrednosti u promenljive
print(x, y, z) # 10 20 30
```

Skupovi (sets)

```
a = {1, 2, 3, 4, 5}
b = {3, 4, 5, 6, 7}
print(a | b)    # unija    -> {1, 2, 3, 4, 5, 6, 7}
print(a & b)    # presek   -> {3, 4, 5}
print(a - b)    # razlika  -> {1, 2}
print(a ^ b)    # xor      -> {1, 2, 6, 7}
```

```
a = {x for x in 'abracadabra' if x not in 'abc'}
print(a)    # {'r', 'd'}
```

Rečnici (dictionaries)

```
room = dict()
room['Aca'] = 331
room['Igor'] = 'M2-1'
print(room)    # {'Aca': 331, 'Igor': 'M2-1'}

for key, value in room.items():    # obilazak recnika
    print(key, value)
```

```
stud = {15123: 'Петар Перић',
        15321: 'Мика Микић'}
ind = int(input('Унесите бр. индекса: '))
if ind in stud:
    print(stud[ind])
else:
    print('Студент није пронађен')
```

Obilazak više listi (*zip* funkcija)

```
indeksi = [15123, 15321]
imena = ["Pera Peric", "Mika Mikic"]
ocene = [{"OOP": 9, "SWE": 8, "CV": 10},
          {"HCI": 8, "SWE": 9}]

# Spisak studenata koji su položili CV
for i, n, o in zip(indeksi, imena, ocene):
    if "CV" in o:
        print(i, n, o["CV"])
# 15123 Pera Peric 10
```

Moduli

- Fajl module.py:

```
def f1():  
    pass  
  
def f2():  
    pass
```

- Fajl main.py:

```
import module  
  
module.f1()  
module.f2()
```

```
import module as m  
  
m.f1()  
m.f2()
```

```
from module import *  
  
f1()  
f2()
```

Paketi

- Grupa PY fajlovi sadržanih u podfolderu čini paket
- `__init__.py` fajl neophodan da bi Python tretirao folder kao paket
- Može biti prazan fajl ili sadržati inicijalizaciju paketa

```
sound/  
    __init__.py  
    formats/  
        __init__.py  
        wavread.py  
        wavwrite.py  
    effects/  
        __init__.py  
        echo.py
```

```
from sound.effects.echo import echofilter  
  
echofilter(input, output, delay=0.7)
```

Rad sa fajlovima

```
with open('proba.txt', 'w') as f:
    f.write('Jedan\n')
    f.write('Dva\n')
    f.write('Tri\n')
```

```
with open('proba.txt') as f:
    content = f.read()
print(content)
```

```
with open('proba.txt') as f:
    lines = f.readlines()
print(lines)  # ['Jedan\n', 'Dva\n', 'Tri']
```

JSON

```
import json

obj = {'key1': 1,
      'key2': 'val2',
      'key3': [1, 'dva', 3.2, False]}
with open('proba.json', 'w') as f:
    json.dump(obj, f)
# proba.json
# {"key1": 1,
#  "key2": "val2",
#  "key3": [1, "dva", 3.2, false]}

with open('proba.json') as f:
    obj2 = json.load(f)
print(obj2)
```


Obrada izuzetaka

```
while True:
    try:
        x = int(input("Unesite celi broj: "))
        break
    except ValueError:
        print("Niste uneli broj. Probajte ponovo...")
print(x)
```

```
try:
    raise NameError('Test greska')
except NameError as ex:
    print('Greska: ', ex.__str__())
    raise
finally:
    print('Kraj')
```

Klasa

```
class Klasa:
    def __init__(self, name): # konstruktor
        self.name = name

    def metod(self, param):
        return self.name + " " + str(param)

class Podklasa(Klasa):
    def __init__(self, name, param):
        super(Podklasa, self).__init__(name)
        self.param = param

    def metod(self):
        return self.name + " " + str(self.param)

obj1 = Klasa('Klasa')
obj2 = Podklasa('Podklasa', 2)

print(obj1.metod(1))
print(obj2.metod())
```

Standardne biblioteke (1)

```
import os
import shutil
import sys
import math

root = os.getcwd() # Vraca tekuci folder
print(root)        # D:\CV_Vezbe\HelloWorld

srcfile = os.path.join(root, 'proba.json')
shutil.copyfile(srcfile, 'D:\\proba.json')

filesAndFolders = os.listdir(root)
print(filesAndFolders)
# ['.idea', 'main.py', 'module.py', 'proba.json', 'proba.txt']
onlyFiles = [f for f in os.listdir(root)
              if os.path.isfile(os.path.join(root, f))]
print(onlyFiles)
# ['main.py', 'module.py', 'proba.json', 'proba.txt']

print(sys.argv)    # ['D:/CV_Vezbe/Hell/main.py']

print(math.cos(math.pi / 4)) # 0.7071067811865476
```

Standardne biblioteke (2)

```
import random
import statistics
import datetime
from urllib.request import urlopen

print(random.random())    # random float iz intervala [0,1)
print(random.randrange(6)) # random integer iz intervala [0,5]
print(random.sample(range(100), 5)) # 5 vrednosti iz intervala [0,99]
print(random.choice(['apple', 'pear', 'banana'])) # random element
print(random.shuffle(['apple', 'pear', 'banana'])) # perturbacija
liste

data = [2.75, 1.75, 1.25, 0.25, 0.5, 1.25, 3.5]
print(statistics.mean(data))    # 1.6071428571428572
print(statistics.median(data))  # 1.25
print(statistics.variance(data)) # 1.3720238095238095

print(datetime.date.today())    # 2019-02-18
print(datetime.datetime.now())  # 2019-02-18 13:49:04.568133

with urlopen('http://tycho.usno.navy.mil/cgi-bin/timer.pl') as
response:
    for line in response:
        line = line.decode('utf-8') # Dekodira binarne podatke u tekst
        print(line, end='')

```

Numpy biblioteka

- <https://docs.scipy.org/doc/numpy/user/quickstart.html>
- Numpy se koristi za rad sa homogenim multidimenzionim nizovima
- Klasa zadužena za Numpy nizove je *ndarray*
- Koristi se za čuvanje slika koje koristi OpenCV
- Dimenzije u Numpy nizu se nazivaju ose (eng. axes)

Atributi *ndarray* objekta

```
import numpy as np

a = np.arange(15).reshape(3, 5)
print(a)
# [[ 0  1  2  3  4]
#   [ 5  6  7  8  9]
#   [10 11 12 13 14]]

print(a.shape)  # (3, 5)
print(a.ndim)   # 2
print(a.dtype.name)  # int32
print(a.itemsize)  # 4
print(a.size)     # 15
print(type(a))    # <class 'numpy.ndarray'>
```

Inicijalizacija Numpy niza

```
import numpy as np

# Na osnovu liste
a = np.array([[1.5, 2, 3], [4, 5, 6]])
print(a)

# Nule (uz specificiranje tipa)
b = np.zeros((3, 4, 2), dtype=np.uint8)
print(b)

# Jedinice (uz specificiranje tipa)
c = np.ones((3, 4, 2), dtype=np.float32)
print(c)

# Uniformna raspodela u opsegu od -1 do 1
d = np.random.uniform(-1, 1, (3, 4))
print(d)

# Gausova raspodela oko vrednosti 0
e = np.random.normal(0, 1, (3, 4))
print(e)
```

Osnovne operacije (1)

```
import numpy as np

a = np.array([1, 2, 3, 4, 5, 6]).reshape((2, 3))
b = np.array([10, 20, 30, 40, 50, 60]).reshape((2, 3))
print(a)
print(b)
c = a + b    # sabiranje po elementima
print(c)
d = a * b    # množenje po elementima
print(d)
e = a @ b.T  # transponovanje i matricno množenje
print(e)
f = a.dot(b.T) # drugi zapis za matricno množenje
print(f)
```


Osnovne operacije (2)

```
import numpy as np

a = np.array([1, 2, 3, 4, 5, 6]).reshape((2, 3))
b = np.array([10, 20, 30, 40, 50, 60]).reshape((2, 3))
print(a)
print(b)
print(a.sum())    # 21
print(a.sum(axis=0))    # [5 7 9]
print(a.sum(axis=-1))    # [6 15]
print(a.sum(axis=(0, 1)))    # 21
print(a.min())    # 1
print(a.max())    # 6
print(np.minimum(a, b))    # [[1 2 3] [4 5 6]]
print(np.maximum(a, b))    # [[10 20 30] [40 50 60]]
```

Promena dimenzije i spajanje

```
import numpy as np

a = np.array([1, 2, 3, 4, 5, 6]).reshape((2, 3))
b = np.array([10, 20, 30, 40, 50, 60]).reshape((2, 3))
print(a)
print(b)
ax = np.expand_dims(a, axis=2)
print(ax)  # [[[1] [2] [3]] [[4] [5] [6]]]
bx = b.reshape((2, 3, 1))  # isti efekat na drugi nacin
print(bx)  # [[[10] [20] [30]] [[40] [50] [60]]]
cx = np.append(ax, bx, axis=2)
print(cx)
# [[[ 1 10] [ 2 20] [ 3 30]]
#   [[ 4 40] [ 5 50] [ 6 60]]]
```

Pristup elementima

```
import numpy as np

a = np.array([1, -1, 2, -2, 3, -3, 4, -4, 5, -5, 6, -6])
a = a.reshape((2, 3, 2))
print(a)
# [[[ 1 -1] [ 2 -2] [ 3 -3]]
#    [[ 4 -4] [ 5 -5] [ 6 -6]]]

print(a[0, 0, 0])      # 1
print(a[0])             # [[ 1 -1] [ 2 -2] [ 3 -3]]
print(a[:, 1])          # [[ 2 -2] [ 5 -5]]
print(a[:, 1, 0])       # [2 5]
print(a[:, 1, 1:])      # [[-2] [-5]]
print(a[0, 0:2, -1])    # [-1 -2]
```

OpenCV

Hello

World 2

```
import numpy as np
import cv2 as cv

img = cv.imread('Lenna.png')

img_r = img.copy()
img_g = img.copy()
img_b = img.copy()

img_r[:, :, 0:2] *= 0
img_g[:, :, 0] *= 0
img_g[:, :, 2] *= 0
img_b[:, :, 1:3] *= 0
imf = img[200:-120, 220:-150]

cv.imshow('Red', img_r)
cv.imshow('Green', img_g)
cv.imshow('Blue', img_b)
cv.imshow('Face', imf)
cv.waitKey(0)

cv.destroyAllWindows()
```

Matplotlib

- Biblioteka koja sadrži različite vrste iscrtavanja matematičkih objekata
- Modul *pyplot* će biti od značaja za prikazivanje slika i grafika.
- Instalacija (slično kao kod instalacije OpenCV-a):
 - Izabrati terminal
 - Zatim otkucati komandu: **pip install matplotlib**

```
import cv2
import matplotlib.pyplot as plt

if __name__ == '__main__':
    lena = cv2.imread("../lena.png")
    plt.imshow(lena)
    plt.show()
```

Figure 1

